

2 Vector Functions

2.3 Curvature and TNB Vectors

2.3.1 Curvature

Definition 2.3.1 A parametrization of a curve $\vec{r}(t)$ is called smooth if $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq \vec{0}$. A curve is called smooth if it has a smooth parametrization.

Definition 2.3.2 The curvature of a curve $\vec{r}(t)$ is defined to be the number

$$\kappa = \left| \frac{d\vec{T}}{ds} \right|$$

Example 2.3.3 Find the curvature of a circle of radius a .

2.3.2 The Tangent, Normal, and Binormal Vectors

Definition 2.3.4 Given a vector valued function $\vec{r}(t)$ the unit tangent vector is given by

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}.$$

Definition 2.3.5 Given a vector valued function $\vec{r}(t)$, the unit normal vector is given by

$$\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}.$$

Definition 2.3.6 Given a vector valued function $\vec{r}(t)$, the binormal vector $\vec{B}(t)$ is given by

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t).$$

Example 2.3.7 Find the unit normal vector and binormal vectors for the function

$$\vec{r}(t) = \langle \cos t, \sin t, t \rangle.$$

2.4 Motion in Space

Example 2.4.1 A moving particle starts at an initial position of $\vec{r}(0) = \langle 1, 0, 0 \rangle$, and with initial velocity $\vec{v}(0) = \langle 1, -1, 1 \rangle$. Its acceleration is given by $\vec{a}(t) = \langle 4t, 6t, 1 \rangle$. Find its velocity and position functions.